



A study on eyelid localization considering image focus for iris recognition

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ABSTRACT

This paper proposes a robust detection algorithm that can be used to detect eyelid region for iris recognition. This research has the following four advantages and contributions compared to the previous works. First, we interpolate the detected eyelashes and specular reflections, which enable us to determine a more accurate searching area of eyelid. Second, we are able to define a limited eyelid searching area by finding the cross position between the eyelids and the outer boundary of the irises. Third, by using eyelid detection mask considering focus value, we can enhance the eyelid detection performance even in the case of defocused iris image. Fourth, by applying the rotation term into the parabolic Hough transform which fits the detected eyelid candidate points, we can detect the accurate eyelid position even in the case of rotated eye. As the experimental results show, the detection accuracy rates were 91.33% and 98.45% when detecting the upper and lower eyelids, respectively.

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1. Introduction

Conventional iris recognition process requires the following three steps. First, it is necessary to localize the pupil and iris regions. Second, the iris feature codes must be extracted. Third, the iris codes must be compared and identified (Wildes, 1997; Daugman, 1993, 2003a,b, 2004). If iris regions are covered by eyelids, recognition accuracy can decrease because the iris feature codes cannot be seen properly. To solve this problem, Miyazawa et al. tried to perform research without using the upper iris region (Miyazawa et al., 2006). Masek proposed an algorithm which divided eyelids from the localized iris regions by using a horizontal line (Masek, 2003), and Liu's method localized eyelids by using two straight lines (Liu et al., 2005). Although all these algorithms have shown good performance in terms of processing time, the recognition accuracy is degraded because eyelids are parabola-shaped in most cases. To solve these problems, in a recent study, Maenpaa proposed an algorithm which detected parabola-shaped eyelids. This algorithm was able to detect eyelid information by using gray values measured in a counter-clockwise direction in the iris regions. If any of the points were shown to have steeply different values than the other points, the point was assumed to be the edge of an eyelid (Maenpaa, 2005). Chen proposed an algorithm that detected the horizontal edges by using the canny edge detector and the longest detected edges that were connected to the others. This algorithm was also able to detect parabola-shaped eyelids (Chen et al., 2006). Also, Cui proposed an algorithm that detected eye-

lashes and located accurate parabola by using the detected eyelash position. This algorithm detected eyelid positions by searching in a y-axis direction (Cui et al., 2004). Jang also proposed an algorithm that detected the eyelid candidate points by using a mask and located eyelid line by using the parabolic Hough transform (Jang et al., 2007).

In general, when a user's iris image is captured by conventional iris camera, it is possible that the eye might be rotated to an extent. So, conventional iris recognition algorithm considers the eye rotation in the case of matching (Daugman, 2003b, 2004). However, the previous researches (Miyazawa et al., 2006; Masek, 2003; Liu et al., 2005; Maenpaa, 2005; Chen et al., 2006; Cui et al., 2004; Jang et al., 2007) did not consider this rotation, which can increase the eyelid detection error. In addition, the previous researches did not consider the blurring of iris image, and the performance of their algorithm is affected by the focus condition. Vezhnevets et al. detected eyelid lines by least square fitting based on the detected eye corner positions. However, they did not consider the blurring of iris image (Vezhnevets et al., 2003).

To solve these problems, this paper proposes a new algorithm that is able to detect eyelid information irrespective of eye rotation and image blurring.

2. Proposed eyelash detection method

2.1. Overview of proposed method

The overview of our proposed algorithm is shown in Fig. 1. First, we detect pupil and iris boundaries, and remove eyelashes and specular reflections. Next, the focus score is measured and the

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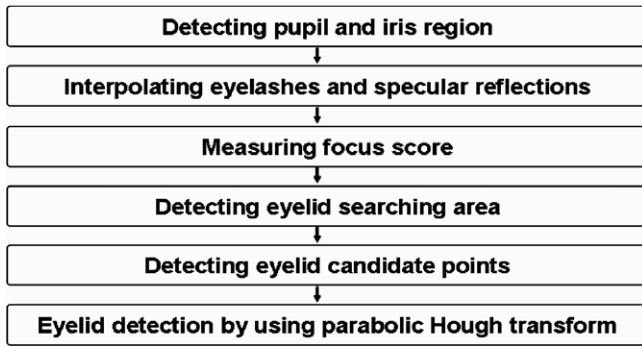


Fig. 1. Proposed eyelid detection algorithm.

eyelid searching areas are detected. After that, we detect the candidate points of the eyelid by using the masks considering focus value in the detected searching area. Finally, we detect the accurate eyelid line by the rotatable parabolic Hough transform based on the detected candidate points of the eyelid. Detail explanations are shown as following.

2.2. Detecting pupil and iris boundaries

We use a two-circular edge detection method considering both pupil and iris information (Daugman, 1993, 2003a,b, 2004; Cho et al., 2006)

2.3. Removing eyelashes and specular reflections

In some cases, eyelid can be occluded by eyelash, which can degrade the detection performance of the eyelid. And the same may occur with the occlusion of specular reflection. To solve these problems, we propose a method of removing eyelashes and specular reflections by using linear interpolation in the x-axis direction. For that, eyelashes were detected by using eyelash detection masks considering focus value (Kang et al., 2005), and the high brightness area (whose gray level was over 230) was detected as specular reflections.

2.4. Focus assessment method

We obtain focus score by measuring mid- and high-frequency amount in an image. And we measure focus score and can discriminate the focused and defocused iris images by using (5×5) pixel sized convolution kernel (Kang et al., 2005).

2.5. Detecting eyelid searching area

In general, to increase the speed and accuracy of eyelid detection, we firstly detect the eyelid searching area as shown in Fig. 2. Eyelids skin generally shows higher reflexivity levels compared to iris region (Yamada et al., 2005). Our proposed algorithm detected four cross points as shown in Fig. 2a and then determined the eyelid searching area, as shown in Fig. 2b.

To detect the cross points of an eyelid, it was first necessary to obtain the gray mean of the iris region which is not occluded by the eyelid area. In general, the horizontal area between iris inner and outer boundaries is not occluded by the eyelid region and we use the gray mean of this area as the threshold for detecting the cross points as shown in Fig. 2a. In a severe case of eyelid occlusion, that area can be hidden by the eyelid, but the visible iris area is too small to be identified.

After obtaining the threshold (gray mean), we calculate the gray difference between two circular regions (having limited circular range) close to the iris outer boundaries. This procedure is iterated in the clockwise direction until the difference exceeds in the threshold. If that is so because the discontinuity exists between iris (or sclera) and eyelid region in general, we determine the point as the right cross point of upper eyelid shown in Fig. 2a. For detecting the remained 3 cross points (the left cross point of upper eyelid and the right and left cross points of the lower eyelid) shown in Fig. 2a, we iterate the same procedure in the clockwise (or counter clockwise) direction. From this, we can detect 4 cross points and determine the eyelid searching area as shown in Fig. 2b (Jang et al., 2007).

2.6. Detecting eyelid candidate points

The formerly proposed algorithms can obtain the eyelid candidate points by using the masks, as shown in Fig. 3a and b in the focused iris image (Jang et al., 2007). But the detection accuracy of eyelid region is degraded in the case of the defocused image, as shown in Fig. 8a. Therefore, we propose a new method of changing the mask adaptively according to the measured focus value. For example, we use the mask shown in Fig. 3c and d in the case of defocused iris image. As shown in Fig. 3c and d, there are 3 separated areas for the kernel components ("1", "0", "-1") and the vertical sizes of each area are increased according to the amount of defocusing. In this case, if the measured focus value by the focus checking mask (as shown in Section 2.4) exceeds in 50, we determine the input image as a focused case, vice versa. The threshold of 50 was determined by the authentication test with CASIA DB (<http://www.nlpr.ia.ac.cn/english/irids/irisdatabase.htm>)

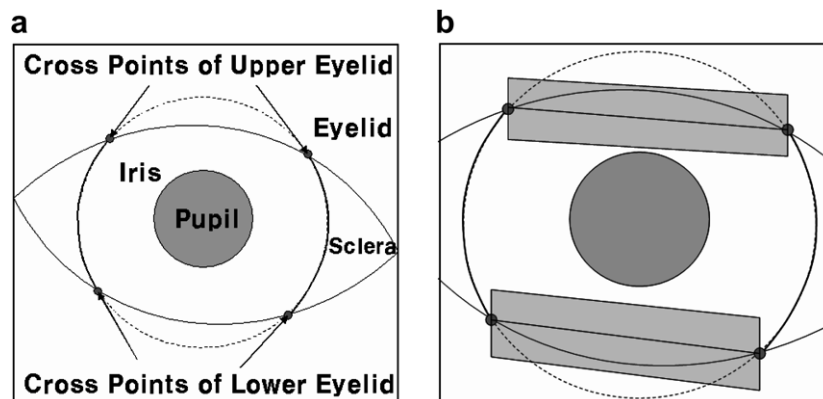


Fig. 2. Detecting eyelid searching area: (a) detecting four cross points (b) determining the eyelid searching area.

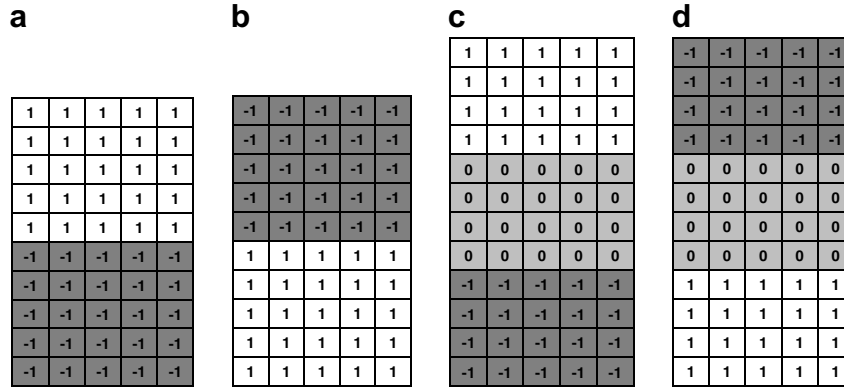


Fig. 3. Masks for detecting eyelid candidate points: (a) mask for upper eyelid in the case of focused image (b) mask for lower eyelid in the case of focused image (c) mask for upper eyelid in the case of defocused image (d) mask for lower eyelid in the case of defocused image.

by Gaussian blurring artificially. That is, if the focus values of genuine iris images exceed in 50, they were authenticated and vice versa.

In the case of the defocused image, the edge border between eyelid and iris is not discrete but somewhat continuous (it means that the gradient value between eyelid and iris region becomes smaller). Therefore, we put the 'zero' points in the mask shown in Fig. 3c and d in order to solve such a problem. Based on the measured focus score, we determine the image focus condition (focused or defocused) and select the mask for eyelid detection as shown in Fig. 3.

In the detected eyelid searching area as shown in Fig. 2b, we detect the eyelid candidate points by using the masks shown in Fig. 3.

Fig. 4 shows the examples of detecting eyelid candidate points with focused and defocused images, respectively.

2.7. Eyelid detection based on rotatable parabolic Hough transform

Based on the detected eyelid candidate points, we located parabolic eyelid lines using Hough transform method. However, conventional Hough transform method takes much processing time (Jain, 1995).

To reduce the processing time, if the measured curvature value is small, we detect eyelids region only by using straight line instead of the parabolic Hough transform as shown in Fig. 5. To measure the curvature value, as shown in Fig. 5, we obtain a straight line ($L(x_0)L(x_n)$) between the detected two cross points of eyelid shown in Fig. 2a. Then, we measure the vertical difference between the straight line and the detected candidate points as shown in Eq. (1). If the measured value exceeds in the predetermined threshold, we perform the parabolic Hough transform. If not, the eyelid is depicted as the straight line to reduce the processing time. The optimal threshold value was determined by the authentication tests with CASIA DB

$$V = \frac{1}{n} \sum_{k=0}^n |P(x_k) - L(x_k)|, \quad (1)$$

where n is the number of the detected candidate points, $P(x_k)$ is the vertical position of the candidate point (x_k) and $L(x_k)$ is that of the point(x_k) on the straight line.

If the curvature value exceeds in the threshold, we perform the parabolic Hough transform as shown in Eq. (2). To consider the eye rotation of user, we apply the rotation factor into the Hough transform

$$E = (\sin \theta(x - h) + \cos \theta(y - k))^2 - a(\cos \theta(x - h) - \sin \theta(y - k)) = 0, \quad (2)$$

where α is the curvature, (h, k) is the vertex point of the parabola, and θ is the rotational angle. To reduce the processing time of Hough transform, we restrict the searching dimensions of four

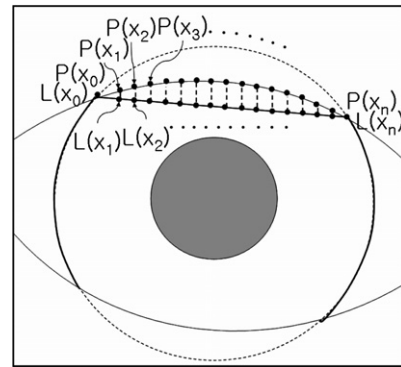


Fig. 5. Measurement of curvature value.

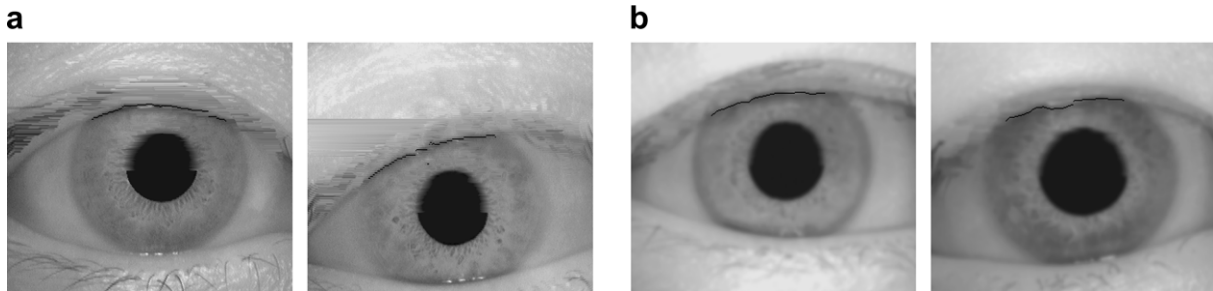


Fig. 4. Examples of detecting eyelid candidate points: (a) with focused image (b) with defocused image.

parameters such as α , (h,k) and θ considering the anthropologic characteristics of human eye.

As shown in Eq. (2), we used the parabolic Hough transform for eyelid fitting. In Eq. (2), α , (h,k) and θ determined the curvature, the translation and the rotation amount of the parabola, respectively. In the case of eyelid rotation, eyelid fitting cannot be successful only by changing α (the curvature value) as shown in Fig. 6b, but it can be done by changing both α (the curvature value) and θ (rotation angle) as shown in Fig. 6c.

Consequently, we used four parameters such as α , (h,k) and θ for eyelid fitting.

3. Experimental results

In this paper, we evaluated the performance of our proposed algorithm by using the CASIA Iris Image Database (version 1.0) (<http://www.nlpr.ia.ac.cn/english/irids/irisdatabase.htm>). The CASIA DB is composed of 756 iris images acquired from 108 different irises of 80 persons, at monthly intervals. Each image in the CASIA DB is an 8-bit gray value with a pixel size of (320*280). The images of 5.2% were acquired from users with contact lenses, but there are no images from users with eyeglasses. The experimental environment was a Pentium-IV 3.0 GHz processor with 512M DRAM.

In Fig. 7, the black curved line denotes the detected eyelid. Similar to Cui's method (Cui et al., 2004), we defined the detection accuracy like following method. That is, if the vertical difference between the detected eyelid point and manually picked one does not exceed in the predetermined margin (we determine the margin as 3 pixels considering the picking error by manually and it was tested by 50 users), we determine the detected point as the correct detected one and vice versa. Consequent detection accuracy was calculated by the ratio between the number of the correct detected points and all the detected ones.

The accuracy of curvature was affected by the detection accuracies of cross points and the candidate eyelid points. When the accuracies of the cross points and the candidate eyelid points were high, the accuracy of curvature detected by Hough transform was also high and vice versa as shown in Table 1.

Table 1

Accuracy of curvature according to the detection accuracies of cross points and candidate eyelid points

Case	Detection of the cross points	Detection of the candidate eyelid points	Detection accuracy of the eyelid curvature (%)	
			Upper	Lower
1	Manually	Manually	99.93	99.71
2	Manually	Automatically	95.76	98.78
3	Automatically	Manually	99.38	98.86
4	Automatically	Automatically	91.33	98.45

In cases 1 and 2, we detected the cross points manually. And in cases 1 and 3, we detected the candidate eyelid points manually in the range of eyelid searching area which was determined by the cross points.

As shown in Table 1, we could know that the detection accuracies of cross points and candidate eyelid points affected that of the eyelid curvature. In addition, we could know that the accuracy of candidate eyelid points affected much more compared to that of cross points.

In the second tests, we measured the detection accuracies of cross points and candidate eyelid points which are shown in Table 2. The accuracies were measured based on the RMS (Root Mean Square) pixel error between the automatically detected points and the manually picked ones.

Experimental results are shown in Table 3, in which the detection accuracy of the upper eyelid was 91.33% and the detection accuracy of the lower eyelid was 98.45%. We compared our proposed algorithm with Cui's algorithm for performance measurement. In Cui's paper, the detection accuracy of the upper eyelid was 97.35% and the detection accuracy of the lower eyelid was 93.39% (Cui et al., 2004). However, for images with a lot of eyelashes, Cui's algorithm was not able to detect eyelids correctly. To prove this, we compared it with our proposed algorithm, using 108 images from the CASIA DB which contained a lot of eyelashes (Table 4). In detail, we detected the eyelashes in the eyelash searching area Kang et al. (2005). If the amount of eyelash

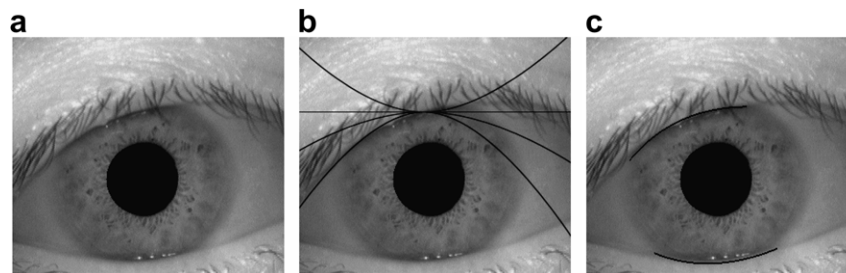


Fig. 6. Eyelid fitting results in the case of eyelid rotation: (a) original image (b) eyelid fitting just by changing the curvature (c) eyelid fitting by changing both the curvature and the rotation angle.

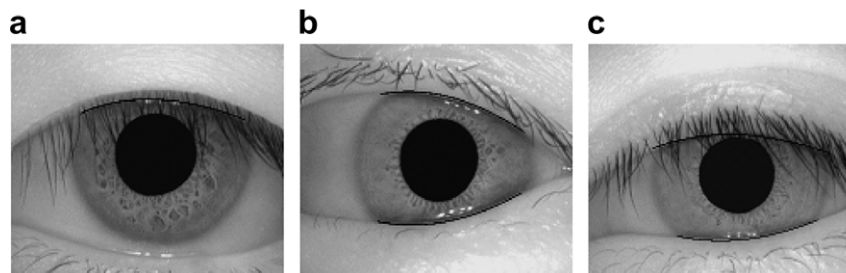


Fig. 7. Detection results of the eyelid line.

Table 2
Detection accuracies of cross points and candidate eyelid points

	Cross points (pixels)	Candidate points (pixels)
Upper eyelid	7.04	2.75
Lower eyelid	5.00	1.29
Total error pixels	6.02	2.02

Table 3
Experimental results of the proposed algorithm using all the iris images of CASIA DB

Case	Accuracy (%)	Processing time (ms)		
		Average	Minimum	Maximum
Upper eyelid	91.33	239	16	542
Lower eyelid	98.45	141	12	506

Table 4
Experimental results only by using 108 iris images having a lot of eyelashes of CASIA DB

Case	Method	Accuracy (%)	Processing time (ms)		
			Average	Minimum	Maximum
Upper eyelid	Cui	82.17	146.1	82.7	224.8
	Proposed	89.99	223	16	530
Lower eyelid	Cui	93.39	616	136.8	1362.4
	Proposed	96.34	159	16	392

exceeded in 50% in the searching area, we determined that it had a lot of eyelashes. Experimental results are shown in Table 4.

Results showed that our proposed algorithm produced higher detection accuracy than Cui's algorithm, as shown in Table 4. In addition, Cui's algorithm required average processing time of 762.1 ms, while our algorithm required that of 382 ms for detecting upper and lower eyelids.

In the Cui's research (Cui et al., 2004), they used the different methods for detecting upper and lower eyelid.

In the case of the upper eyelid detection, first, they used the frequency information to detect eyelash. And then, they used the

Table 5
Detection accuracy (Jang et al., 2007) according to the amount of defocus of CASIA DB

Case	Accuracy (%)			
	σ of 0.0	σ of 1.0	σ of 2.0	σ of 3.0
Upper eyelid	91.33	90.69	81.37	75.32
Lower eyelid	98.45	96.35	95.47	95.43

Table 6
Detection accuracy of proposed method according to the amount of defocus of CASIA DB

Case	Accuracy (%)			
	σ of 0.0	σ of 1.0	σ of 2.0	σ of 3.0
Upper eyelid	91.33	90.82	90.75	90.72
Lower eyelid	98.45	96.45	96.41	96.42

Table 7
Iris recognition accuracy with and without eyelid detection (using the CASIA DB)

Accuracy without eyelid detection		Accuracy with eyelid detection	
d	EER (%)	d	EER (%)
4.449	1.331	6.864	0.239

detected eyelash information to determine the eyelid shape. Finally, they determined the y-position of the eyelid shape by moving eyelid shape in vertical direction and measuring the change of the difference value.

In the case of the lower eyelid detection, however, they used the mean and variance information in the detected iris region in order to separate the eyelid region from the iris region. And then, they detected the edge points to determine the eyelid shape. Finally, they determined the y-position of the eyelid shape by moving the eyelid shape in vertical direction and measuring the change of difference value.

In Cui's research (Cui et al., 2004), the processing time of detecting upper eyelid was much faster than that of the lower one and

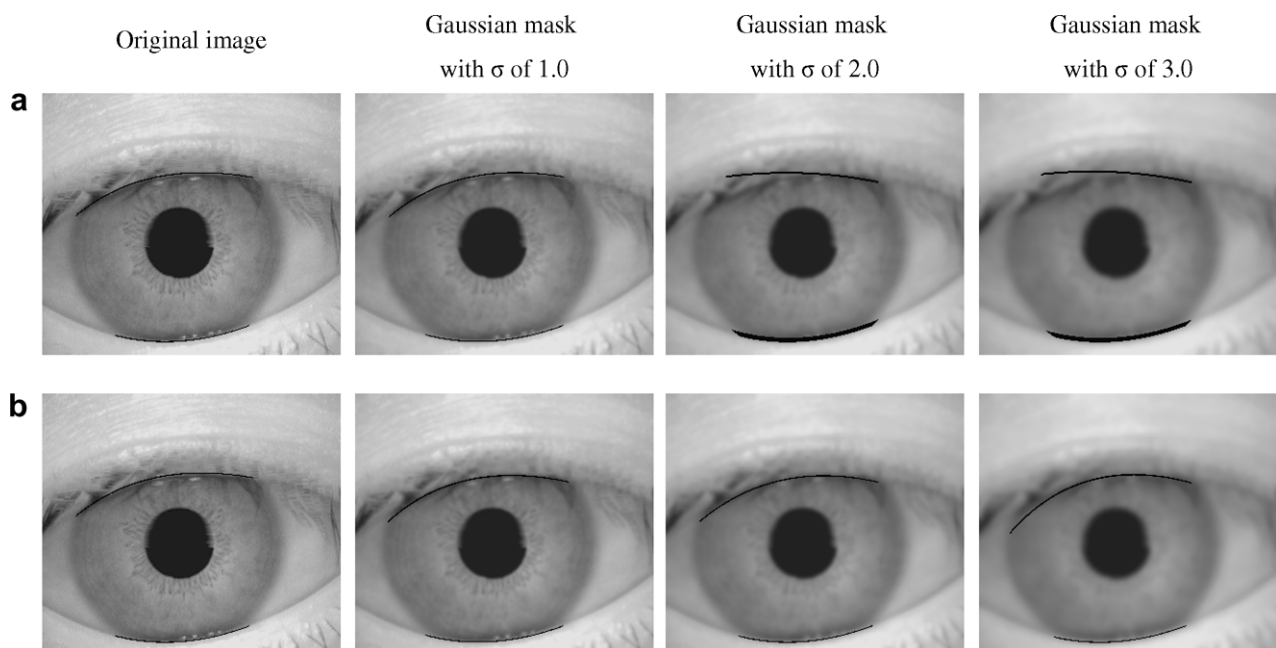


Fig. 8. Detection results of the eyelid line with the focused and defocused images: (a) detection result in the case of using a mask not considering focus value (b) detection result in the case of using a mask considering focus value.

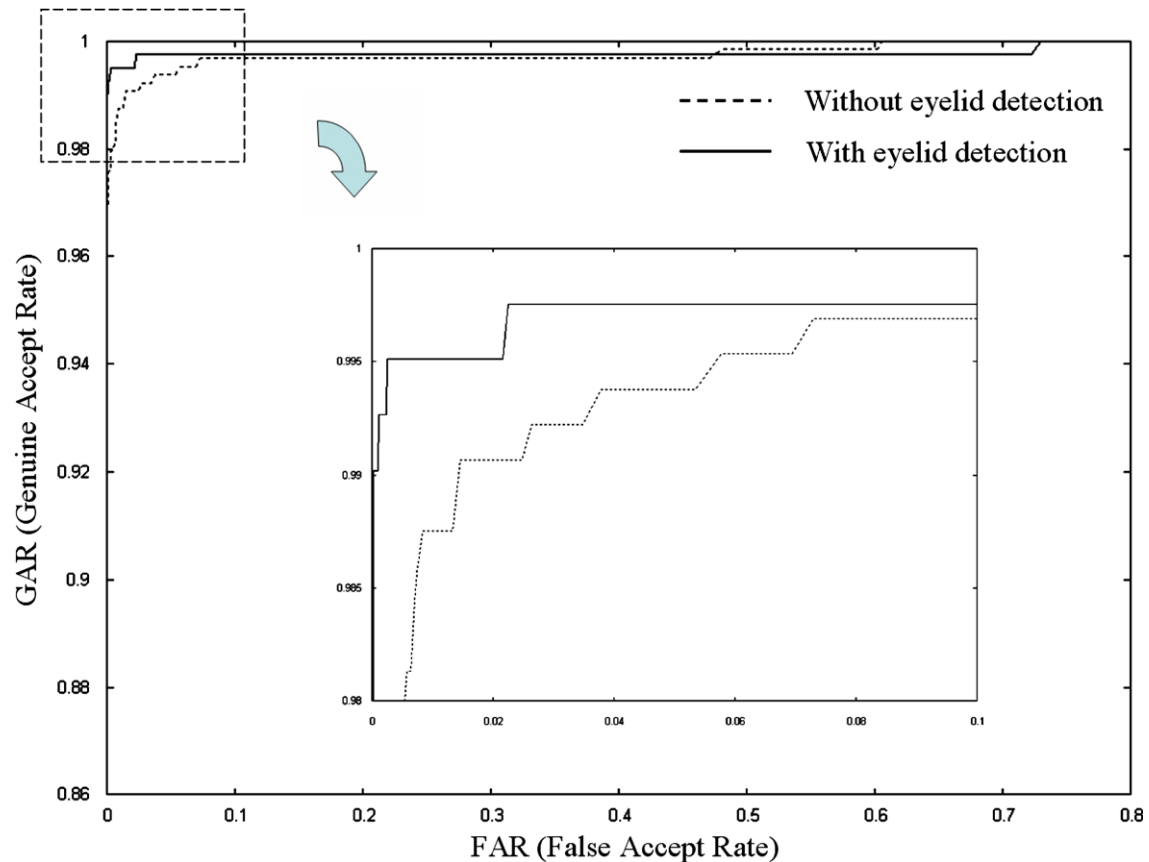


Fig. 9. ROC curves.

they did not explain the reason. But, according to our experiment, we could know that because there was much error about the separation of the lower eyelid from the iris region only by the mean and variance information, many edge points were generated in the separated low eyelid searching region. Consequently, there existed too much edge candidate points for measuring the change of the difference value of the lower eyelid shape and it increased the processing time much.

In the next experiments, we measured the detection accuracy according to the amount of defocusing. For that, we made the images of CASIA DB blurred by Gaussian smoothing mask with the incremental of σ . As shown in Fig. 8, in the case of defocusing, our method could detect the eyelid line more accurately compared to the previous method (Jang et al., 2007).

According to Tables 5 and 6, we can know that the performance of proposed method is not affected by the amount of defocusing. In the next test, we also measured iris recognition accuracy with and without the proposed algorithm. For iris code generation and matching, we used a conventional 2D Gabor filter (with the optimal kernel size of 25 and frequency of 0.05) and the Hamming distance (Daugman, 2004). The kernel size and frequency were selected in order to minimize the EER (Equal Error Rate) when working with the CASIA DB (Park et al., 2007a,b). As experimental results show, accuracy was much enhanced when using the proposed method, as shown in Table 7 and Fig. 9.

4. Conclusion

This paper proposes a robust detection algorithm that can be used to detect eyelid region for iris recognition irrespective of eye rotation and image defocusing. In future work, we plan to re-

duce the processing time and the parameter estimation of Hough transform. And we will have more field test with non-ideal iris images.

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